



#### 4.2.4 Wrap-Up: Reflection

Complete each statement as you reflect on today's learning.

1. The concept that made the most sense to me today is...
  
  
  
  
  
  
  
  
  
  
2. One concept that I still do not understand from today is...

## Unit 4, Lesson 3: Identifying Key Features of Lines From Tables and Equations



### Benchmark

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.



### Learning Targets

- ☐ I can identify key features of a linear function from a graph.
- ☐ I can identify key features of a linear function given a table or an equation.



### 4.3.1 Warm-Up: Equations of Lines

1. Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$y = -\frac{\boxed{\phantom{00}}}{\boxed{\phantom{00}}}x + \boxed{\phantom{00}} \text{ is the same line as}$$
$$\boxed{\phantom{00}}x + \boxed{\phantom{00}}y = \boxed{\phantom{00}}$$



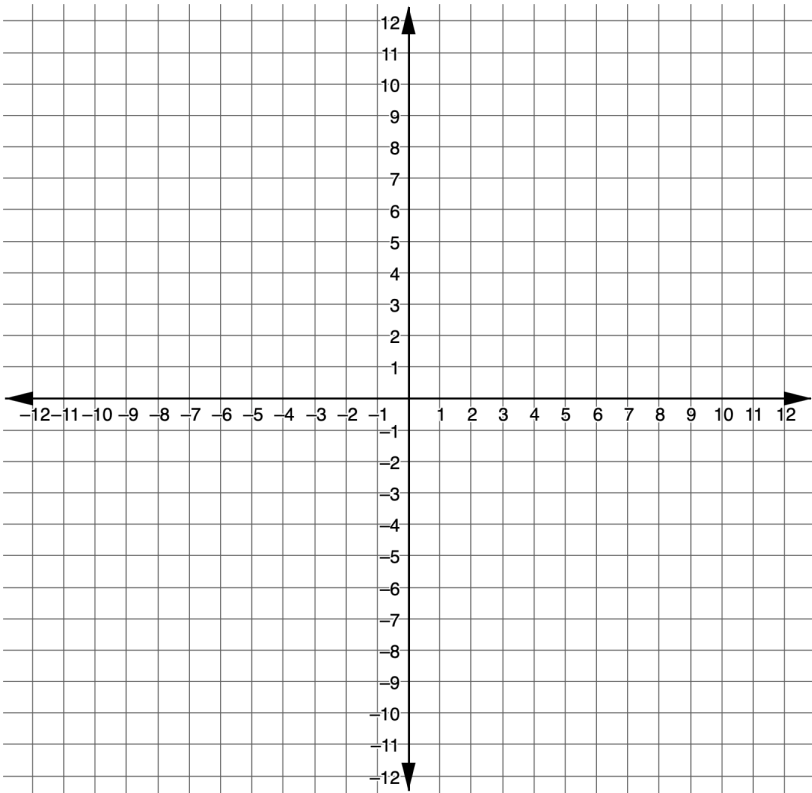
### 4.3.2 Collaboration: Key Features of a Linear Function From a Table

1. A table of values that represents a function is shown.

| $x$ | $y$  |
|-----|------|
| -2  | -9   |
| -1  | -6.6 |
| 0   | -4   |
| 1   | -1.5 |
| 2   | 1    |
| 3   | 3.5  |

- a. With your partner, discuss which key features of this function, if any, can be determined from the table of values.
  - ☐ Domain
  - ☐ Range
  - ☐ Slope
  - ☐ Increasing/Decreasing
  - ☐  $x$ -intercept
  - ☐  $y$ -intercept

b. Use the table of values to graph the function it represents,  $f(x) = \frac{5}{2}x - 4$ .



c. Find the key features of the function  $f(x) = \frac{5}{2}x - 4$ .

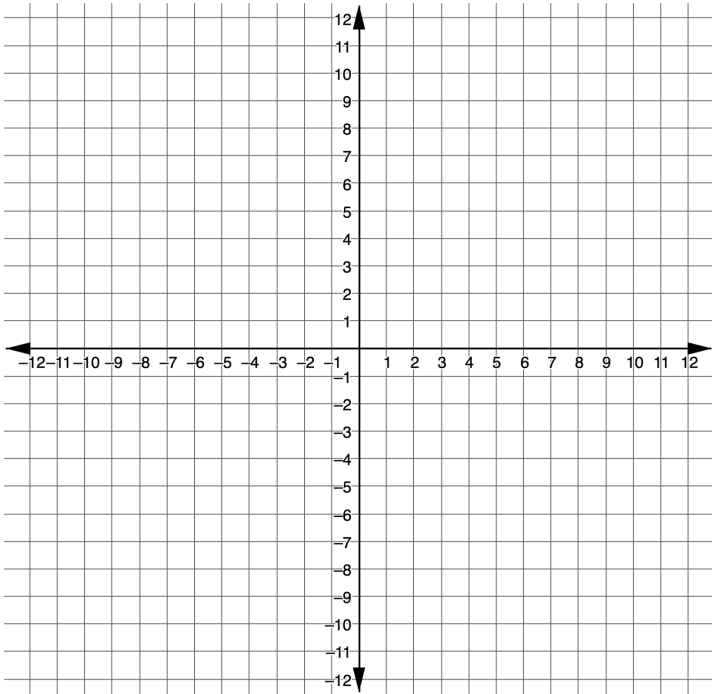
| Domain | Range          | As $x$ -values increase, $y$ -values ... |
|--------|----------------|------------------------------------------|
|        |                |                                          |
| Slope  | $x$ -intercept | $y$ -intercept                           |
|        |                |                                          |



### 4.3.3 Guided Instruction: Different Forms of a Line

1. Consider the equation  $y + 3 = -\frac{3}{4}(x + 8)$ .

a. Graph  $y + 3 = -\frac{3}{4}(x + 8)$ .



b. Find the key features of  $y + 3 = -\frac{3}{4}(x + 8)$ .

| Domain | Range          | As $x$ -values increase, $y$ -values ... |
|--------|----------------|------------------------------------------|
|        |                |                                          |
| Slope  | $x$ -intercept | $y$ -intercept                           |
|        |                |                                          |



2. The equation  $y + 3 = -\frac{3}{4}(x + 8)$  is just one way of writing an equation or a function that represents the line you graphed. The table shows the four common representations of the same line.

| Point-Slope Form              | Slope-Intercept Form       |
|-------------------------------|----------------------------|
| $y + 3 = -\frac{3}{4}(x + 8)$ | $y = -\frac{3}{4}x - 9$    |
| Standard Form                 | Function Notation          |
| $3x + 4y = -36$               | $f(x) = -\frac{3}{4}x - 9$ |

- a. Which forms of the line listed are the easiest to find the slope?
- b. Which forms of the line listed are the easiest to find the  $y$ -intercept?



### 4.3.4 Your Turn!

1. A group of students are discussing how to find the slope and  $y$ -intercept of the line with the equation  $9x - 15y = -75$ .
- a. Use the description each student gives to complete their work.

| Malcolm                                                                                                                                                      | Olivia                                                                                                                                                                                                                      | William                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To find the slope and $y$ -intercept, I rewrite the equation to slope-intercept form.                                                                        | To find slope, I mentally move the $x$ -term to the other side and divide by the coefficient of the $y$ -term. To find the $y$ -intercept, I divide the value the equation is equal to by the coefficient of the $y$ -term. | To find slope, I write the coefficient of the $x$ -term divided by the coefficient of the $y$ -term. To find the $y$ -intercept, I write the value the equation is equal to divided by the coefficient of the $y$ -term. |
| Malcolm's Work                                                                                                                                               | Olivia's Work                                                                                                                                                                                                               | William's Work                                                                                                                                                                                                           |
| $9x - 15y = -75$<br>$-15y = -9x - 75$<br>$\frac{-15}{-15}y = \frac{-9}{-15}x - \frac{75}{-15}$<br>$y = \underline{\hspace{1cm}}x + \underline{\hspace{1cm}}$ | Slope = $\frac{\boxed{\hspace{1cm}}}{-15}$ or $\boxed{\hspace{1cm}}$<br><br>$y$ -intercept = $\frac{\boxed{\hspace{1cm}}}{-15}$<br><br>or $\boxed{\hspace{1cm}}$                                                            | Slope = $\frac{\boxed{\hspace{1cm}}}{-15}$ or $\boxed{\hspace{1cm}}$<br><br>$y$ -intercept = $\frac{\boxed{\hspace{1cm}}}{-15}$<br><br>or $\boxed{\hspace{1cm}}$                                                         |

- b. Which student would NOT find the correct slope and/or  $y$ -intercept?



### 4.3.5 Wrap-Up: Reflection

Color in the arrow up to the statement that best describes your current understanding of identifying key features of a linear function.



I'm so confident, I can explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and listened, but I am finding this challenging. I will make sure I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

## Unit 4, Lesson 4: Relating Key Features of Lines to Real-World Contexts



### Benchmark

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.



### Learning Targets

- ☐ I can locate the key features of linear functions using graphs, tables, equations, and written descriptions.
- ☐ I can determine if the domain and range are viable for the context.
- ☐ I can interpret the slope and  $y$ -intercept based on the context and determine if the interpretations are viable.